

Markscheme

Mathematics: analysis and approaches

Practice question bank

25. (a)

Consider the function $f(x) = \frac{5x^2 - 19x + 20}{(x-1)(x-2)(x-3)}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$.

attempt to write $A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$ (M1)

$$5x^2 - 19x + 20 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2) \quad \text{A1}$$

$$\text{substituting } x = 1: 6 = 2A \Rightarrow A = 3 \quad \text{A1}$$

$$\text{substituting } x = 2: 2 = -B \Rightarrow B = -2 \quad \text{A1}$$

$$\text{substituting } x = 3: 8 = 2C \Rightarrow C = 4 \quad \text{A1}$$

[5 marks]

(b)

Hence find the exact value of $\int_4^5 f(x) dx$.

$$\int f(x) dx = 3 \ln |x-1| - 2 \ln |x-2| + 4 \ln |x-3| \quad \text{(M1)}$$

$$\text{evaluating between } x = 4 \text{ and } x = 5 \text{ gives } \ln\left(\frac{4096}{243}\right) \approx 2.825 \quad \text{A1}$$

[2 marks]